# 8 Demonstration and Definition: Aristotle's Positive Views in *Posterior Analytics* B.8–10 and B.16–18

## 8.1 Introduction

Aristotle needed to find a method to establish which features of a kind define it. The definitional features, thus secured, should be (in some way) prior and sufficient to underwrite the unity of the kind. His method should essentially involve the use of demonstration. In this way, he could satisfy the Simple Condition and Definitional Constraint, which he employed in his criticisms of the methods of general deduction and division.

In the context set by B.3–7, Aristotle's aim was even more precisely defined. He had to achieve these goals in a way which was not vulnerable to the Formal and Independence Problems (as set out in Chapter 3). Thus, he had to show how demonstration could be of use in the definitional task, even though the relevant demonstrations did not have as their conclusions definitions of the kind.[334]

Part of Aristotle's solution has already been considered.[335] The conclusions of the relevant syllogisms need not contain both *definiens* and *definiendum,* provided that they are systematically connected to the term to be defined. The relevant link is provided by the Stage 1 account of what the term signifies. Thus, if 'thunder' signifies the same as 'a certain type of noise in the clouds', the demonstrative conclusion:

Noise belongs to the clouds

is connected in an obvious way to the term to be defined. For, as suggested above, Aristotle's accounts of what the term signifies provide the

[334] The phrase 'definitional task' is intended to be neutral at this point as between two options: the task of searching for definitions, and the task of laying out definitions once discovered. Demonstrations cannot contain both the term defined and its definition in the conclusion.

[335] See Chs. 2 and 3 above.

springboard for an investigation into the existence and nature of the kind.[336]

The stage is now set for Aristotle's attempt to satisfy his own Simple Condition and Definitional Constraint. I shall first consider his discussion in B.8, $93^{a}16$–$^{b}14$. Investigation of the demonstrations used there (Section 8.2) and in B.16–17 (Section 8.3) will enable us to understand part of his strategy for connecting definition and explanation in *Posterior Analytics* B. But the resulting picture is more complicated than it might at first seem (Sections 8.4 and 8.5).

## 8.2 Demonstration, General Deduction, and Efficient Causation: The Basic Case

Aristotle offers two examples of syllogisms in B.8 to illustrate the connection between the practice of definition and of demonstration. They run as follows:

<table>
<tr><td rowspan="2">[EXAMPLE 1]</td><td>Being eclipsed belongs to all things being screened by the earth. Being screened belongs to the moon (or perhaps to all moons of kind K).</td></tr>
<tr><td>Being eclipsed belongs to the moon (all moons of kind K).[337] ($93^{a}30$–1)</td></tr>
<tr><td rowspan="2">[EXAMPLE 2]</td><td>Thunder/Noise belongs to all fire-quenchings. Fire-quenching belongs to the clouds (all clouds of kind K).</td></tr>
<tr><td>Noise belongs to the clouds (all clouds of kind K). ($93^{b}9$–12)</td></tr>
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[336] More precisely, they fulfil two roles:(a)They give the basis for an appropriately articulated two-term statement which is the starting point for a search for the relevant middle term (which explains why the two terms are so connected).and(b) They indicate an as yet not fully specified type of (e.g.) noise, which can be specified further at the second and third stage of enquiry.

[337] The precise logical form of these syllogisms has been the subject of much discussion. I assume (for present purposes) that they are in the Barbara form, and contain no singular terms. Thus, I prefer to construe reference to the moon as to (e.g.) the moon in certain stages of its career.

In these examples, although an effect is demonstrated, the conclusions do not contain definitions.[338] Even when the term being defined (e.g. 'eclipse') appears in the conclusion, the latter is not a definition of eclipse (or even of eclipse of the moon).

Why is this method preferable to that involving general deductions? In these demonstrations, the middle term (e.g. 'fire-quenching') specifies the efficient cause of:

(1) an eclipse of the moon

and

(3) noise occurring in the clouds

We can answer the 'Why?' question by giving the efficient cause of the phenomenon specified in the conclusion.

The causal order displayed in these demonstrations is relevant to the issue of priority encapsulated in the Simple Condition. In the case of thunder, the quenching of fire is causally more basic than the occurrence of noise in the clouds, because the former is the efficient cause of the latter.[339] This asymmetry is captured in syllogism (2), but not in the syllogisms of the method of general deduction. For, in the syllogism:

Noise in the clouds belongs def to all fire-quenchings.
Fire-quenching belongs def to all thunder.
Noise in the clouds belongs def to all thunder.

the middle term is not the efficient cause of noise in the clouds belonging to thunder. Indeed, this syllogism gives no ground for treating the second

[338] In this syllogism, I take the conclusion reached at 93$^b$10 f. *not* to be of the form:Thunder is noise in the clouds.It should be noted, however, that in B.10, 94$^a$7–9, Aristotle might appear to take this to be the conclusion 'of the demonstration of the what it is'. Is Aristotle's view inconsistent? Perhaps Aristotle is not over-concerned with the precise nature of the syllogism involved. (On these issues, see J. L. Ackrill, 'Aristotle's Theory of Definition' in Berti (ed.), *Aristotle on Science,* 359–63.)However, in B.10, 94$^a$8 'this' can refer either to the phrase 'noise in the clouds' or to 'thunder is noise in the clouds'. The former seems preferable because it keeps the terms of the syllogism the same as those specified in 94$^a$4–7:'noise', 'fire being quenched', 'the clouds'. To suppose otherwise is to deprive 'the clouds' of the status as a separate term which it clearly has in 94$^a$4 ff. and 93$^b$9 ff.There remains a problem concerning the substitution of 'noise' and 'thunder' in 93$^b$9 ff. But this is a different issue, related to the permissible substitution of names and (parts of) accounts of what names signify.

[339] In this Chapter, I use interchangeably as translations for '*aition*', 'cause', 'causal explanation', and 'explanation'. This issue is discussed in more detail in Chapter 10.

premiss as prior to the conclusion. By contrast, through the practice of demonstration we grasp that a feature is definitionally prior because it is prior in the order of efficient causation. The quenching of fire is the start of a causal process which culminates in noise occurring in the clouds.

Why, for Aristotle, should our definitional practices depend on our way of explaining phenomena? It is not merely that we *know* that certain features are definitionally prior because they are causally prior (B.8, $93^a3$–5). Rather, as he remarks in *Posterior Analytics* A.2, $71^b31$:

> things are prior (i.e. by nature) since they are causes.

The order of definitional priority is determined by the order of causation. Fire being quenched is definitionally prior to noise in the clouds because it is causally prior. It is the efficient cause. What it is to be something (as captured in the definition) and the basic relevant cause are one and the same (B.2, $90^a14$–15).[340] The dependence of our practice of definition on that of explanation reflects a metaphysical interconnection between essence and causation.[341]

We were prepared for a connection between defining and explaining by Aristotle's remarks in *Posterior Analytics* B.6. There, he suggested that there can be no account of particular definitions in terms independent of demonstration. Since demonstration captures the order of causation, there can be no account of particular definitions which is independent of the causal order.[342] In giving demonstrations we track, in the case of a kind, the metaphysical order of causation back to one cause of all its relevant necessary features. Had the order of causation not been connected in this way to the theory of definition, it would be (at least, epistemically) possible for there to be an alternative route to find what is definitionally prior. For, if definitional priority were not determined by causal priority, it would be (in principle) possible for us to come to know the former by a route independent of the latter. Aristotle, as we noted above, does not seriously consider the possibility of an alternative epistemic route of this type. So, his preference for the method of demonstration is, it appears, based on his assumption that the order of definition is metaphysically dependent on the order of causation. This is why we need to engage in demonstration to establish correct definitions.[343]

340 Equally, since 'thunder' is taken to be equivalent in meaning to 'noise in the clouds', the quenching of fire is also definitionally prior to thunder.

341 I use the terms 'practice of defining' and 'practice of explaining' to separate what we do in giving definitions/explanations from the subject matter of such definitions and explanations: essences and *explananda* / *explanantia* (or causes/effects).

342 See Ch. 7.

343 It may be objected that Aristotle's discussion of *nous* in the final chapter of the *Analytics* shows a route to grasp definitions which is independent of the theory of demonstration. I shall argue in Chapter 10 that this is not so. However, the present focus is restricted to the preceding chapters of *Post. An.* B.

Aristotle's remarks elsewhere suggest the same view. In B.16 ($98^{b}19$–21) he contrasts the syllogism which demonstrates the cause:

> Things that are near do not twinkle.
> The planets are near.
> The planets do not twinkle.

with one which demonstrates the fact:

> Things that do not twinkle are near.
> The planets do not twinkle.
> The planets are near.

Planets fail to twinkle (according to Aristotle) because they are near. They are not near because they fail to twinkle (A.13, $78^{a}37$ f.). The order of causation is fixed independently of the order of definition. However, he remarks in B.16 that:

> It is clear that being screened by the earth (B) is the cause of the eclipse (A) and not vice versa: for, B is specified in the account of A, so that it is obvious that the latter is known via the former and not vice versa. ($98^{b}21$–4)

The order of efficient causation is *clear* because it is captured in the order of definition. But this, as we can now see, is because the order of definition is determined by the order of causation. The stars' position is definitionally prior to their failure to twinkle because the former is the efficient cause of the latter.[344] Here, too, the interdependence of our practices of definition and explanation reflects a metaphysical connection between essence and *explanans*/cause.

Other features of Aristotle's method rest on the same view. His 'immediate propositions' are a case in point (see B.8, $93^{a}36$). We reach, in giving explanations, an immediate proposition when there is no further cause which connects (e.g.):

(1) deprivation of light and the earth screening ($93^{a}30$–6)

[344] In B.16, $98^{b}22$ ff. Aristotle writes:That the eclipse is not the cause of its being in between but the latter of the eclipse is clear; for, being in between is in the definition of eclipse, so that the latter is known by means of the former and not vice versa.This might appear to take the order of definition to explain the order of causality (in opposition to the thesis developed above). However, Aristotle notes in this passage *not* that A causes B because A is used to define B, but rather that it is *obvious* that A causes B because A is used to define B. Obviousness is an epistemic notion, which reflects the order of our knowledge. But this is fully consistent with the line of interpretation developed to understand B.8–10, which concerns the metaphysical order of definition.

or

(2) noise and fire being quenched.

If there were a further cause, our investigation would be incomplete until we discovered it ($93^b12$–14). Immediate propositions are bedrock from a causal point of view. They describe a direct, unmediated, causal connection. No further cause can be invoked to account for the connection they express.[345] In B.8 two features are central to the examples of demonstration:

(1) In discovering the answer to the 'Why?' question, we trace the pattern and order of efficient causation.
(2) In discovering the answer to the 'Why?' question, we find at the end of our enquiry an immediate proposition. At this point, there is no further causal relation to be discovered relevant to the effect in question.

In both examples, the relevant causal pattern is one of efficient causation. By following this, we come to know the essence of the phenomenon in question. Here, what is definitionally prior (viz. the essence) is determined by what is causally prior.

A further example of this order of determination is suggested by Aristotle's identification of *the* essence with *the* cause (B.2 $90^a14$–15, see also the epistemic version in B.8 $93^a3$–4). If one traces back the relevant causal line from the effect, one will find *one* feature (*the* cause) which explains why (in the case of thunder) noise occurs in the clouds. This feature will also explain why thunder possesses its other necessary properties: why it is accompanied by lightning (for example), or why it is noisy. For, the presence of these properties will be explained because the fire, when quenched, produces noise and lightning. In this way, there is *one* efficient cause which brings about all the other necessary properties of thunder. This efficient cause would be discovered whichever of the relevant necessary properties

345 In these examples of immediate propositions, we find two distinct phenomena which are immediately related. The cause in these cases is other than the phenomenon caused ($93^b$ 19, cf. $93^a$ 7), as *fire being quenched* is other than *noise*. As such, immediate propositions of this type are different in kind from the cases discussed in B.9, where Aristotle is concerned ($93^b$ 21–5) with primary objects which are identical with their causes, or essences. It is crucial to separate immediate or primary *objects* (which are identical with their essences ($93^b$ 21–5)) from immediate *propositions* which state unmediated (causal) connections between distinct phenomena. (There may be a class of immediate propositions which are concerned with the relation between primary objects and their essences. If so, they will be a subclass of immediate propositions, whose other species will involve cases where the kinds or objects are distinct from their causes, but are non-mediately connected with them.) These two sets of cases are distinct, and confusion results if they are conflated. For a contrasting view, see R. Bolton's paper 'Definition and Scientific Method in Aristotle's *Posterior Analytics* and the *Generation of Animals* ', in A. Gotthelf and J. Lennox (eds.), *Philosophical Issues in Aristotle's Biology* (Cambridge, 1987), 138–42.

had been the starting point of one's enquiry. For, it is the common cause of all the relevant features. Thus, the essence is the one cause of all the kind's derived necessary properties.

Aristotle, armed with the method of demonstration, can satisfy the requirements of his own Simple Condition. Thunder is a unity because there is *one* common efficient cause which explains the presence of its necessary properties. It is because fire is quenched that there is noise in the clouds, noise accompanied by lightning, etc. The presence of this prior feature underwrites the unity of the kind. Similarly in the case of man. Aristotle had asked: why do all the following properties belong to man:

footed, two-footed animal (92$^{a}$30)

and

featherless, footed, two-footed, mortal animal (92$^{a}$1)?

He now has an answer: because there is one common cause which explains man's possession of all of these properties. This one cause underwrites the unity of the kind. The relevant cause will be captured by the middle term of a syllogism in which each proposition says one thing of one thing non-accidentally (93$^{b}$36 f.; cf. 72$^{a}$9). By contrast, there is no one cause which explains why something is both musical and literate. This is an accidental unity because the feature which explains one property fails to explain the other.

Aristotle's model of enquiry encapsulates a metaphysical picture. Enquiries of this type are successful when there is one common cause which is prior in the way specified to the other features of the kind. In the simple cases Aristotle discusses, the definitionally prior feature is the efficient cause. More generally, the prior feature will be the common cause in whatever causal mode is appropriate for the kind in question. The presence of this one feature will provide the basis for the unity of the kind.

Aristotle's model allows him to satisfy the requirements of the Definitional Constraint. Certain features will be relevant to definition if they possess the type of priority and unity required of good definition. They will possess the latter because they are causally prior and the one common cause of all the other relevant features of the kind. Such features will be mentioned in our definitions because they are causally prior unities of this type. Other features may be included in our definitions if they are immediate consequences of the one common cause. (For further discussion, see Chapter 9.)

These suggestions allow Aristotle to make substantial progress in the *Analytics*. His model satisfies the unity and priority requirements which he used to criticize rival proposals in B.5–6. Had he failed to meet them, his criticisms of those methods would have lost much of their force. The reasonable conclusion might have been that Aristotle's own requirements could not be met. Perhaps he should have jettisoned them and retained either the method of general deduction or division.

The grounds for Aristotle's preference for the method of demonstration have now become clear. He can use it to establish particular definitions because it will show which features are the causally basic, unifying features of the kind. Features cannot be known to be prior in this way unless they are seen as the basis for demonstrative explanation of the appropriate type. Without this, there would be no way to know which features are definitional. As Aristotle comments in his concluding remarks in B.8:

> In cases where the cause is different from the effect, it is not possible to know what it is without demonstration, but there is no demonstration *of it*.(93^b18 f.)

Through giving demonstrations, we can use the notions of causal priority and of one unified cause to capture what is definitionally prior and to underwrite the unity of the *definiendum*. Further, the success of this method is non-accidental, since its epistemic virtues reflect the metaphysical dependence of definition on causation. In these respects, it is clearly preferable to the method of general deduction.

Aristotle's basic model has many gaps. In the cases so far considered, he has focused on examples involving efficient causation. Can this model be extended to cases involving other types of cause (see Sect. 8.5)? Nor have we understood other central aspects of the model. Why are certain features taken as the *explananda?* What constrains the appropriate causal connections? However, despite these gaps, we can reach an interim conclusion: In Aristotle's account causal priority (of some form) determines (in some measure) what is definitionally prior. In this way, our practice of definition rests on, and is partially determined by, our practice of explanation.

In the next Section, I shall investigate further the type of causal explanation which Aristotle employs in these contexts. It will soon become apparent that the connection between defining and explaining is more complicated than has so far been suggested. For, it turns out that it is appropriate to use certain forms of causal explanation because they are the right ones to yield definitions.

## 8.3 Explanation and Essence: B.16–18: The Basic Model Extended and Refined

In *Posterior Analytics* B.16–18 Aristotle discusses further the type of causal explanation to be used in arriving at definitions (see B.17, 99^a21 ff.). Examination of these passages may shed more light on the procedure he employed to satisfy the Definitional Constraint. For, several aspects were left obscure in B.8–10.[346]

In B.16–18 Aristotle considers two questions about causal explanation:

(A) If the *explanandum* is present, is the *explanans* always present?
(B) If the *explanans* is present, is the *explanandum* always present?

In Aristotle's own terminology, the relevant question is as follows:

(A*) If there is an eclipse, is there always a screening by the earth?
(B*) If there is a screening by the earth, is there always an eclipse?

As he puts it, do the relevant terms ('eclipse', 'interposition of the moon') convert? We might say: is the occurrence of the *explanans* (considered as a type) both necessary and sufficient for the occurrence of the *explanandum* (as a type)? (cf. $98^a35–^b4$)

Aristotle's answers are cautious but revealing. One of his examples is biological. He asks:

> Why do trees shed their leaves? If this is because of solidification of the sap, then if a tree sheds its leaves, solidification must belong to it. And if solidification occurs, in a tree, *not just in anything you like,* it must shed its leaves. ($98^b36–8$)

Aristotle's requirement that the terms involved be equal in extension and convert cannot apply to all the cases of causal explanation he considers.[347] For, when the Athenians come to be at war with the Persians because they are the first to attack (B.11, $94^b1$ f.), the two terms cited ('being at war' and 'being the first to attack') do not convert. Since a war of this type needs at least two opponents, the aggressors and those who respond to aggression, not all who are at war can be the aggressors.[348] More generally, there are many different ways to come to be at war. So, the question is acute: Why is conversion *required* in the particular examples cited in B.16?

It is not immediately clear that the terms cited in the example above ('sap solidification', 'leaf loss') do convert. Do trees (or even broad-leaved trees) lose their leaves *only* when their sap solidifies? Aristotle is raising exactly this question when he asks:

[346] Thus, it is not clear whether all or any of the terms convert.

[347] Barnes argues powerfully that conversion is not the rule in all demonstrations (*Aristotle's Posterior Analytics,* 258–9). The substantial issue is whether there is any non-trivial group of demonstrations which require conversion.

[348] Another example of a causal explanation which does not meet this stringent requirement is given in *Post. An.* B.17, $99^b$ 4–5 in his discussion of the causes of longevity in different species.

> is it true that if the *explanandum* (e.g. plants shedding their leaves) holds, the *explanans* also holds (e.g. their sap solidifying)? (98ᵃ36–8; see 98ᵇ30–2)

Surely there can be different explanations of why such trees lose their leaves. Perhaps, as Aristotle comments:

> It is necessary that something explanatory is present, but not that everything which might be a cause is present. (98ᵇ30–1)

In some cases, sap solidification might lead to leaf loss, in others the use of pesticide or the presence of disease (as there might be various explanations of why people are at war).[349]

There is a related question about the sufficiency of the proposed explanations. Why is it that if trees' sap solidifies, they must shed their leaves? Surely we can explain why some trees shed their leaves by saying:

> it is because their sap solidifies,

without thinking that in every case when a tree's sap solidifies it will lose its leaves?

Aristotle has offered an answer to these questions given (in part) in the words already cited:

> if solidification occurs, in a tree, *not just in anything you like,* it must shed its leaves. (98ᵇ36–8)

His concern is to explain why (e.g. broad-leaved) trees *as such* and *universally* (98ᵇ34) lose their leaves. He aims to find an explanation of why this happens to all such trees solely in virtue of their being such trees. Hence, if their sap could solidify without their leaves falling, one would not have explained *universally* why trees of this kind *as such* lose their leaves. One might, of course, have explained why these trees in certain conditions (e.g. when grown near a chemical plant) lose their leaves, but one could not explain why *all* broad-leaved trees *as such* do so.[350]

This strategy can be used to defend Aristotle's view that this type of explanation should provide necessary conditions for the occurrence of the effect. He is aiming at giving an explanation of the following form:

> In *all* members of one kind K, a feature F which belongs to them *as such* always has one and only one cause C.

[349] Equally, there could be a case where trees shed their leaves from differing causes on differing occasions: e.g. in some cases because of chemical pollution, in others because of sap solidification. One cannot avoid this problem solely by limiting the cases in question to those involving trees (as Barnes suggests, *Aristotle's Posterior Analytics,* 253).

[350] Contrast the case of the Athenians coming to be at war with the Persians. There might well be no explanation of why the Athenians came to be at war, solely in virtue of their being Athenians. One would typically need to refer to something they had done (e.g. supported the Milesians), and not just to their nature.

In all broad-leaved trees, leaf loss (if it belongs to them all in virtue of their being broad-leaved trees) will always and only occur because of sap solidification. Explanations of this form will fail if some such trees' sap can solidify without their shedding their leaves. For, then, sap solidification alone will not account for their leaf loss. Equally, they will fail if some of their leaves can fall without their sap solidifying. Explanations of the desired form will apply to all actual and possible cases of broad-leaved trees losing their leaves. Otherwise, they cannot apply to *all* these trees *solely in virtue of their being such trees,* but only to such trees in certain conditions.

Aristotle clarifies his favoured explanatory structure further in B.17, 99ᵃ23–9. He notes that 'shedding leaves' applies to all vines, but that it extends further to figs and other types of tree. Thus, he writes:

> Shedding leaves [viz. B term] holds universally of all figs [D term]—for, I call *universal* that with which they do not convert, and *primitive universal* that with which severally they do not convert, but with which, taken together, they do convert and have the same extension. (99ᵃ32–5)

In this case, shedding leaves will be the *primitive* universal which applies to all and only trees of a given type (e.g. broad-leaved trees) which comprise the class of figs, vines, etc. What needs to be explained is why:

> Shedding leaves belongs to all and only broad-leaved trees.

The latter is the *primitive* universal to which leaf-shedding belongs. As before, all and only broad-leaved trees will shed their leaves, and leaf-shedding will belong to such trees *as such* (*Post. An.* A.4, 73ᵇ26 ff.). In this case, it appears to be in the nature of leaf-shedding to belong to broad-leafed trees and no others. That is, the type of leaf-shedding at issue is (partially) defined as something that happens to trees of that kind.[351] What governs the selection of the relevant primitive universals is that they satisfy the conversion requirements in this distinctive way. That is, the *explananda* are convertible and *per se* connected. (What is to be explained is why the type of leaf shedding found in these trees belongs to these trees as such.) There need be no such relation between being the Athenians and being at war.

If leaf-shedding belongs to all broad-leaved trees *per se,* the explanatory feature should also belong to such trees in this way. For, if this were not so, one would not have explained why leaf loss (of the relevant type) belongs to these and only these trees (solely in virtue of their being these trees). Thus, it must be in the nature of sap solidification (of the relevant type) to belong to trees of this type, and only to such trees. In explanations

[351] This will be a case of a *per se* 2 predication: i.e. one where it is in the nature of A to belong to B. By contrast, *per se* 1 predications are those where it is in the nature of B's that A's belong to them.

where the feature to be explained is one which belongs to the kind *per se,* the *explanans* must also belong to that kind in the same way. So, the *explanans* and *explanandum* must convert, because both are to be understood as belonging *per se* to the same kind. In this way, the explanations at issue are able to meet the convertibility requirements.

This form of explanation of why leaf-shedding of this type occurs in broad-leaved trees will invoke the fundamental essence of this type of leaf-shedding.[352] The explanatory syllogism will be the basis of an account of what this type of shedding leaves is. As Aristotle says:

> What is shedding leaves? The solidifying of sap at the connection of the seed. (99ª28 f.)

The explanatory syllogism in question would run as follows:

Shedding leaves Φ all and only solidifiers of sap.
(E) Solidifying of sap Φ all and only broad-leaved trees.
Shedding leaves Φ all and only broad-leaved trees.

Explanations of this form, which contain *primitive* universals in both the conclusion and the premisses, will provide the basis for a full definition of the relevant phenomenon as the type of occurrence in broad-leaved trees which is brought on by their sap solidifying. They are structure-revealing explanations precisely because their conclusion contains a *per se* connection of this type. I shall call explanations with this feature *structural explanations.*

It is no accident that when we find structural explanations of this type we discover commensurate universals required for definition. Provided that the *explanandum* is a unitary property whose definition adverts to one kind to which it belongs primitively, an appropriate *explanans* will provide us with a basic definition of that property.[353] Aristotle's favoured mode of explanation is designed to allow us to come to know which features define the property in question. For, his explanations take as *explananda* features

[352] It might also be in the nature of broad-leafed trees to shed their leaves. In that case, there would be a *per se* 1 predication as well. However, this is not required in this context. One *per se* connection is enough for Aristotle's purposes.

[353] This claim is only that if a predicate belongs to its subject *primitively,* it is coextensive with it. This does not entail the far stronger claim (favoured by Zabarella in *Opera Logica* (Frankfurt, 1608) ) that if a predicate belongs to two subjects non-primitively, it must belong to a common kind which embraces these two subjects primitively. Some predicates may *not* belong primitively to one proper subject at all. At best, like longevity, they may belong only to a conjunction of such subjects. Zabarella's proposal is effectively criticized by Barnes (in *Aristotle's Posterior Analytics,* 258), who concludes that *primitiveness* plays no role in generating commensurate universals. If my weaker construal of primitiveness is correct, primitive universals will be commensurate, and *explananda* of this sort will lead to definitions.

and kinds (such as primitive universals) which are specified in such a way as to ensure their convertibility.

Nor need this method be confined to the definition of properties. If the feature to be explained is one which is used to differentiate the kind, the explanation will advert to the basic essence of the kind. Explanations of this type may result in definitions either of the feature or the kind. This is because the *explananda* are either the *differentiae* of a kind or a property which is itself defined as belonging to that kind. In the former case one will arrive at the essence of the kind, in the latter at the definition of the feature which belongs to the kind.

If we find structural explanations we will uncover definitions because their *explananda* are *per se* connected (in the ways indicated). It appears that, at this point, the practice of explanation depends on that of definition. For the *per se* connections the former invokes rest on definitional connections between feature and kind. Indeed, this is what distinguishes this form of explanation from the type involved in explaining why the Athenians come to be at war with the Persians.

This appearance is, I shall argue, confirmed by Aristotle's more detailed discussion of this form of explanation in the remainder of B.16 and B.17.

## 8.4 The Basic Model Amplified*

The importance of the principle that:

> In one kind, one feature F of the appropriate type has one and only one cause C

emerges in Aristotle's discussion of a variety of cases which appear to threaten it. In the first group, leaf-shedding turns out not to be a unified phenomenon because there are two different explanations of why it can occur in what is one kind. Here, we should distinguish between natural and diseased leaf-shedding. If we find ‘two accounts or more [of the phenomenon], it is clear that what we are seeking is not a single thing but several’. For example, if intolerance of insult and indifference to fortune have a different account, there will be two types (*eide*) of pride ($97^b$13–25). The absence of one feature jointly necessary and sufficient to explain all cases of pride would show that there are in fact two types of pride. Here, Aristotle accepts a principle of the following form:

P(1) If objects $k_i$ . . .$k_n$ are the same in species and there is a different explanation of why they possess a given property, there is more than one property to be explained.

Aristotle maintains the principle of conversion of cause and effect by separating various types of effect.

In a second group of cases, leaf-shedding might turn out not to belong *per se* to just one kind, such as figs. It belongs instead *per se* to a wider kind which involves figs and other trees as well. Aristotle gives a mathematical as well as a biological example.[354] Here, his strategy is to find a wider kind to which the relevant feature primitively belongs *per se*. In this way, he maintains his basic thesis that for one kind there is one explanation of the relevant phenomenon by introducing a wider kind to which the feature belongs *per se*.

The second type of case has the following structure:

A Φ all and only B.
B Φ all D.
A Φ all D.

Here A and B (leaf-shedders and broad-leafed trees) are coextensive, but D (fig trees) is not coextensive with either. Aristotle's strategy is to find a group D* (sap solidifiers) to which D and other species belong, which is coextensive with A and B. This background group (D*) is the one which appears in the explanations not of why D's are A, but of why D and (e.g.) F's are A. (See B.18, $99^b$7–8.)[355]

In both the first and second groups discussed, Aristotle shows how structural explanations generate definitions of the relevant properties. There is, however, a third group of cases which has a different structure. It runs as follows:

A Φ all B.
B Φ all and only D.
A Φ all D.

[354] It runs as follows:(1)Having external angles equal to four right angles applies to all triangles.Here, the feature to be explained applies more widely than to the class of triangles. Aristotle's response is to argue that having external angles equal to four right angles applies to a larger group consisting of triangles, quadrangles, and perhaps others. Thus, we should say (according to Aristotle):(2)Having external angles equal to four right angles applies to group G consisting of triangles, quadrangles . . .Group G, which consists of triangles, quadrangles, etc., is co-extensive with the things that have external angles equal to four right-angles. Thus, the middle term which explains why (B) is true will also be co-extensive with this larger group. For, the latter is the group to which the property of having an angle sum equal to four right angles applies *as such*.

[355] This pattern is also to be found in the biological cases of leaf-shedding mentioned above.

Here, B is the cause of A's being D, but not of everything which is D. Aristotle introduces the case as follows:

> But [356] indeed in some cases B is the cause of D's being A [with emphasis on the D]. In such cases A must extend further than B. If it did not, what reason is there for B to be the cause of this [viz. of A's belonging to all D]? ($99^{a}$35–7)

His line of thought appears to fall into several distinct stages:

(1) Take a case in which B (a middle term) is the cause of D's being A ($99^{a}$35 f.).
(2) This case is problematic, let us suppose, because A extends further than B. (If this were not so, B would not be the cause of D's being A but of some wider group (e.g. D's and E's) being A.[357] ($99^{a}$37).

Aristotle seeks to analyse this problematic case as follows:

(3) If A extends beyond D [e.g. to all E's and F's], and its presence beyond D is not explained by E's being B, then E's must be a group distinct from D [viz. apart from the group unified by feature B]. ($99^{a}$37–9).

If (3) were not so, and Es were not a group distinct from Ds, we would not be able to say:

A Φ all E.

while denying that:

E Φ all A. ($99^{a}$39–$^{b}$1)

For, if E's formed the same group as D's (viz the E* group) and:

A Φ all E*s.

were true, there would be commensurate universals of the form:

A Φ all E*'s (where E and D form this one unified group).

[356] I read in $99^{a}$ 35 ff., 'τoιoταϛ δὲ ἠδD' (or 'τoιϛ δὲ') rather than 'τoιϛ δὴ' as in the MSS, and take this to introduce a second case (distinct from that discussed in $99^{a}$ 30–5, where all the terms convert when the appropriate group is specified). The second case introduced in $99^{a}$ 36 cannot have this feature since the A and B terms do not convert. There must, therefore, be a break between these two passages. Since in the MSS 'δε' and 'δη' are both written in the same way, there is no better evidence for the present text than for my emendation. On the standard reading nothing is added by saying that B is the cause of D's being A, since this is already implicit in $99^{a}$ 30–3.

[357] I translate '*ti māllon aition estai touto ekeinou* ' as 'why is this [viz. B] rather the cause of that [viz. D's being A]?', and understand the implied contrast to be 'rather than B's being the cause of something other than D's being A'. The contrast should not be with 'rather than D's being A being the cause of B'. As Ross (*Aristotle's Prior and Posterior Analytics,* 673) and Barnes (*Aristotle's Posterior Analytics,* 258 ff.) note, Aristotle has several good reasons to rule out the latter possibility.

and there would be one explanation of why A belongs to all D and all E ($99^b1$ f.). But if E is (as we shall assume) a unified group distinct from D (with its own separate explanation), there will not be commensurate universals of the form:

A Φ all and only E*

because D and E do not form a unified group. Here, there is no primitive universal E* to be introduced, as D and E do not form a genuine kind.

Aristotle is careful to add one important rider:

> But will E be some one thing? We must enquire into this. Let it be C. ($99^b3$ f.)

This caveat prevents one from concluding that E must be a unified group simply because it plays the specified role in explanation. E might turn out not to be a unified kind, but one which requires further division. Alternatively E might fall below the level of unified kinds. Aristotle's preferred form of explanation (of *per se* features via necessary and sufficient conditions) is itself constrained by the demand that the feature to be explained belong to a genuine unity.[358]

In this type of case, there is one property to be explained (e.g. longevity), but the explanation is different for different animals (e.g. quadrupeds and birds). The absence of one feature necessary and sufficient to explain the presence of this property in all cases forces us to distinguish between the kinds in question. There can indeed be, as Aristotle says:

> several explanations of the same thing, but not *for things of the same sort.*($99^b4$–6)

There will be differences between the kinds of object under investigation in all cases where the same property is present but its presence is differently explained.

In this case, Aristotle commits himself to a further principle:

P(2) If the same property belongs to objects $k_i$ . . .$k_n$ but its presence is differently explained, objects $k_i$ . . .$k_n$ must differ in species (or in genus).

This thesis is also a reflection of his fundamental principle that for one kind K there is one and only one explanation of one genuine feature F. Here, this principle is used to differentiate kinds and not features. If the explanation is different for:

358 There is no need to interpret these lines as indicating 'Aristotle's awareness of the weakness of the preceding lines' (as Barnes suggests in his note on $99^a$ 38, *Aristotle's Posterior Analytics,* 257.) They show rather the complexity of the conditions he required of the appropriate explanation.

A Φ all E.

and

A Φ all D.

the kinds involved must be distinct ($99^b2$–4).

In the third group of cases, one would not arrive at a definition of longevity, because it does not belong *per se* to one kind. Cause and effect fail to convert. There is no one element which explains longevity in kinds D and E. Explanations of this type cannot reveal the structure or definition of longevity. Nor need they generate definitions of the varying kinds (D and E), unless the relevant feature happens to be among the *differentiae* of these kinds. Although acceptance of P(2) forces one to separate and eventually define kinds when there are two explanations of the presence of one property, it need not lead us directly to a definition of either kind or feature. We do not automatically have in the third case a structural explanation of either the property or the kind.

This leads us to focus on a crucial point. The *explananda* in structural explanations are ones which belong to all the relevant kinds *per se.* Either the property is defined in terms of its belonging to a given kind, or the kind is defined in terms of its possessing certain properties. This type of explanation is distinguished (in part) by the fact that its *explananda* are definitionally connected. The *explananda* have to be of this type if they are to fit the convertibility schemata required if we are to discover essences by this route. When this is not so (as in the third case), we will no longer arrive at definitions of either property or kind. We need to think of (e.g.) leaf-shedding as leaf-shedding-in-these-trees for it to be the type of primitive universal required as an *explanandum.* In this way, Aristotle's favoured type of structural explanation appears to rest on *explananda* provided by our practices of definition (via their concern with the notion of primitive universals).

## 8.5 The Interdependency of Defining and Explaining

The interconnections between defining and explaining in Aristotle's account are close and intricate. In 8.2, we saw him using explanatory concerns to make determinate the intuitions concerning priority and unity which are involved in giving definitions. However, the story in 8.3 and 8.4 is subtly different. Here, his relevant form of structural explanation appears to rest on resources drawn from the practice of definition. For, the latter plays a role in providing the relevant *explananda* and in underwriting the convertibility of the terms required in structural explanations.

However, the arguments of 8.3 and 8.4 are incomplete. Aristotle needs to make clear *how* the practice of definition selects various features as the appropriate *explananda*. For, without that addition, some might interpret Aristotle as follows: In relying on definitional connections in arriving at the relevant *explananda* we are merely using results which are, at basis, dependent on our practice of explanation. No doubt, we can rely (epistemically) on definitional connections in looking for the *explananda*. But what makes these connections definitional is their (e.g.) being the immediate consequence of one underlying common cause. Similarly, it may be said, convertibility should be seen as the consequence of the presence of this type of causal structure. It need not be motivated by independent non-explanatory definitional resources. Perhaps our practice of definition rests throughout on that of explanation (without reciprocal support).

In the next Chapter, I shall seek to make good this lacuna. The relevant *explananda* are (in at least some cases) determined by aspects of the theory of definition, independent of the requirements of explanation. But to see this one needs a fuller grasp on the role of definition in B.13–15.

There are, nonetheless, further reasons (prior to B.13 ff.) to think that, for Aristotle, constraints implicit in our practice of definition are at work in determining the relevant pattern of explanation. Such constraints are not confined to the *explananda,* but also underlie some of the requirements placed on the *explanans.* Consider cases where there are several causal stories which explain the same phenomenon: efficient, teleological, grounding, etc. (B.11, 94$^{b}$27 f.). Here, Aristotle suggests that there can be efficient and teleological explanations of the occurrence of thunder (94$^{b}$31–4), and notes that this problem arises for a variety of natural phenomena (94$^{b}$34–6). What makes one of these causal stories the right one to yield the definition of the kind or feature in question?

Since Aristotle identifies the answer to the 'What is it?' and the 'Why?' questions (B.2, 90$^{a}$15), it is natural to think that only one of these causal stories gives us knowledge of the appropriate type of what the kind is. Definitions aim to give us knowledge of what the kind or substance is (see B.3, 90$^{b}$16, *Topics* Z.4, 141$^{a}$27 ff.). In the case of thunder, the efficient cause (fire being quenched in the clouds) makes it fully intelligible to us why thunder has the other genuine (or *per se*) features it has: being a noise, being accompanied by lightning, occurring under certain atmospheric conditions. The first two are consequences of fire being quenched, the third points to the surrounding circumstances required for fire to be quenched. By contrast, the teleological cause cited (to threaten those in Hades (94$^{b}$33–4)) would not make it clear why thunder occurs in the clouds or in certain atmospheric conditions, even if it explained why thunder was noisy. Only one of these explanations makes the nature of the kind fully intelligible to us in the way required to answer the 'What is F?' question.

This thesis can be generalized. In Aristotle's view, the efficient cause gives the basic essence of some phenomena, the teleological cause of others (*Meta*. Z.17, $1041^a28$–30). This can be true even if there is always one type of efficient and one type of teleological cause for the same phenomenon. Thus, anger may always be brought about by (e.g.) an insult, and aim at revenge (*De An*. A.1, $403^a31$ ff.). The latter constitutes the essence of anger, since it is this which makes the phenomenon fully intelligible to us. Once we understand that anger aims at revenge, we see that it has to be brought about by something which the angry person wishes to repay (e.g. an insult). This is because revenge has to involve reacting adversely to some wrong done. By contrast, if one knew only that a person was in a state brought about by an insult one would not thereby know that the person was looking for revenge. In order to know that, one would also need to know some further facts about their psychology. Only the teleological cause is sufficient by itself to make the phenomenon of anger fully intelligible to us.

A similar story can be told about the nature of (e.g.) houses, where there is one efficient cause (the art of house-building) and one teleological cause (safety for goods, etc.), each of which by itself can explain (in some way) all the relevant features of a house. However, only the teleological cause tells us 'what a house is'. Why? Not solely because it is a common cause of all the relevant features of the house.[359] For, so too (*ex hypothesi*) is the efficient cause. There must be further constraints at work beyond those generated by the practice of explanation alone.

In the case of the house, the further element could be introduced as follows. If one had only the efficient causal story, something would remain unclear: why the art of building is the kind of art it is. One would merely take for granted (as an unexplained datum) why this art functions as it does. The teleological account is preferable because it shows why the efficient cause is as it is. Only one of these causal stories makes the nature of the phenomenon fully intelligible to us in the way required to answer the 'What is F?' question. Here, background assumptions about the requirements of good definition select which causal story is the relevant one for definitional purposes.[360]

Elsewhere, Aristotle appears to represent our practice of explanation as dependent on that of definition. Consider his contention that there cannot be explanatory chains of infinite length (A.19, $82^a7$ f).[361] He argues for this as follows: If there were such chains, one could not arrive at knowledge of what the kinds are or of their definitions (A.22, $82^b37$–$83^a1$). Rather, definitions must be of finite length if we are to grasp them (see also $83^b4$–8). This argument is, at first glance, weak. Why would it follow from the fact

[359] See, e.g. *Meta*. H.2, $1043^a$ 15 ff.

[360] For further discussion of this point, see Ch. 10.

[361] I am indebted at this point to discussion with Henry Mendell.

that definitions cannot be of infinite length that demonstrations cannot be? What prevents there being explanations of infinite length which do not begin with definitions? Why should definition play this central role in the practice of demonstration?

The argument of the previous paragraph suggests an answer to these questions. Demonstrations, as Aristotle conceives them, must begin with statements which concern the essences of the kinds in question. For, this is a constraint on what demonstrating is. To demonstrate is to give an argument which begins with a statement about a causally basic (non-demonstrable) feature which can answer the 'What is F?' question. Thus, the demonstratively basic statement must be finite in length; for, otherwise it could not make the phenomenon fully intelligible to us. If demonstrations are to meet this requirement, they must begin with definitions which reveal the relevant essences.

This argument is revealing. Aristotle does not attempt to show that every science must contain a set of basic propositions which are underivable from the remainder, and then argue that these propositions must be definitions which can satisfactorily answer the 'What is F?' question. That is, he does not impose a structural constraint on demonstration, and argue that only a subclass of knowable propositions can meet it.[362] Rather, he begins with the substantial claim that demonstrations must begin with statements concerning essences, which (if they are to answer satisfactorily the 'What is F?' question) must be knowable by us. If so, the practice of demonstration is once again constrained by considerations drawn from our practice as definers.

These requirements serve to make the relevant type of structural explanation more determinate. The cause and effect convert, and both should belong *per se* to the kind. The cause cited must provide the basis for an answer to the relevant 'What is F?' question. That is, there must be one non-complex feature which is the common cause of (e.g.) the kind's other *per se* properties. Further, the nature of the common cause should not

[362] For a contrasting view, see Robin Smith's paper 'Immediate Propositions and Aristotle's Proof Theory', *Ancient Philosophy* 6 (1986), 47–55. Smith argues convincingly that Aristotle's arguments in *Post. An.* A.19–22 cannot depend on the claim that every regress of propositions must end in a basic set of premisses which are self-evident or epistemically immediate. For, Aristotle claims only that the starting points are definitions which reveal essences, and makes no reference to the epistemic notion of self-evidence in this passage. However, Smith concludes from this that the relevant starting points must merely be ones not deducible from any other true proposition, and then notes that there is no obvious step from this claim to any which concern (non-demonstrative) knowledge of definitions. On my view, the starting points are *non-demonstrable* because they are not the causal consequences of any further claim. Also, if they are to be *starting points of demonstration,* they must (a) be non-demonstrable and (b) state the relevant basic essence. If the relevant essences are the objects of non-demonstrative knowledge (as emerges in *Post. An.* B.8–10), the starting points of demonstration will themselves be capable of being known non-demonstratively.

stand in need of any further explanation. Rather, it should serve to make the whole phenomenon fully intelligible to us. These latter requirements follow from the demand that the relevant basic cause be used as the basis of the definition of the phenomenon.

Essences, in this account, are the starting points of demonstration and answer the 'What is F?' question. This conception of essence (and of definition) explains other features of Aristotle's explanatory practices. Thus, he insists that demonstrations must begin with *one* common cause and not a conjunction or disjunction of separate causes. These claims are natural if what is sought is the essence of the kind, a single cause which can provide the basis for the 'What is F?' question. Without this assumption, there would be nothing amiss with finding several separate causes for each of a kind's *per se* features. If we aim at definition, we will not be content with uncovering a conjunctive cause. For, that would fail to make all aspects of the kind transparent. In particular, it would not explain why all its features are interconnected in the way they are.

If this is correct, Aristotle's overall strategy can be formulated in terms of a mutual dependence between our practices of definition and of demonstration. While some of his claims about defining follow from his views about explaining, the constraints he imposes on structural explanation are (in some measure) grounded in the practice of definition. Each of these practices is incomplete without resources drawn from the other.

If this general picture is correct, Aristotle's account of the interdependency of the practices of definition and of demonstration involves a further thesis: the co-determination of essence and causation. Essences play a central role in a certain style of causal explanation, but the relevant type of causal story itself involves essences. This metaphysical thesis grounds Aristotle's epistemological claim that in knowing the answer to the 'Why?' question we know what the kind in question is. We achieve knowledge of essences by tracing back a certain pattern of explanation to its roots in a particular type of cause. But the relevant type of explanatory structure is one in which essences must be the basic *explanantia*. Essences and structural causation stand or fall together.

## 8.6 Definition and Explanation: Interdependency and Its Problems

The interdependency of defining and explaining, introduced in the last Section, offers a way to understand Aristotle's claim in B.2 that the answer to the 'Why?' and to the 'What is it?' questions are one and the same. Why does he hold (at least) the following:

- [F is the essence/basic definer of A ↔ F is the fundamental explanatory feature in a given type of explanation]?

Aristotle (I have argued) requires that the type of explanation be a structural one, concerned with (e.g.) features which belong to the kind *per se*. We are now in a position to be able to understand his strategy more fully.

The initial claim in B.2 could be understood in a number of ways. On one view, the left-hand side of the bi-conditional, and in particular the notion of essence, is taken as basic. Here, the key metaphysical concept is that of one unified essence which constitutes the nature of a kind. This essence will indeed be the basis of a given pattern of explanation, but it can be understood as the essence in terms independent of its role in explanation. Successful explanations may invoke essences, but this is because such essences are the ontological building blocks on which good explanations rest.

There is a radical alternative to the first viewpoint. Here, one seeks to construct an account of what essence is from features which play a certain role in explanation. On this view, essences are to be defined in terms of their pivotal role in a form of explanation, which can be understood independently of the notion of essence. In this account, the right- and not the left-hand side of their bi-conditional is taken as basic. To be an essence is to play a given role in explanation.

We have seen reason to doubt that Aristotle held either of these alternatives. In his view, each side of the bi-conditional depends on the other. In his unified picture of defining and explaining, there are (at least) three key features (in cases where the cause is different from the effect).

## (1) The Immediacy Condition

In searching for the definition of a phenomenon one reaches an immediate proposition, where no further cause can be invoked to explain the connection between the phenomenon and the feature invoked. This is causal bedrock. The causal feature invoked in the immediate proposition is causally prior to the features to be explained.

## (2) The Unity Condition

The feature invoked in (1) must itself be one common cause of the other necessary features of the phenomenon which are explained.

## (3) The Explanatory Condition

The feature invoked in (1) and (2) must be necessary and sufficient to explain why the phenomenon *as such* has the other properties it possesses *as such*.

These constraints arise from the practice of offering structural explanations. But, at crucial points, such explanations involve features drawn from our practices of definition. Thus, the relevant explanatorily basic feature is selected because it satisfies additional definitional constraints. The appropriate explanatory picture cannot be completed without the assistance of features drawn from the theory of definition.[363]

There is a further feature of Aristotle's account. In it, some kinds possess certain of their features in virtue of their belonging to wider-ranging kinds. The sub-kinds will be made what they are by their possessing distinctive unifying explanatory features which account for their possession of their distinctive necessary properties. Equally, the kinds to which they belong will be made what they are in the same way. Thus, both genera and species will be essentially defined by reference to that feature which is central in the relevant type of explanation of their possession of their other necessary properties. Thus, in addition to theses (1)–(3) above, Aristotle is committed to a further claim:

## (4) The Hierarchical Conception

Kinds may possess features in virtue of their belonging to wider-ranging kinds of which they are sub-kinds. The definitions of the higher-order kinds will themselves be fixed in ways which meet (1)–(3).

These four features form the basis of Aristotle's picture of defining, explaining, and classifying as set out in these chapters of the *Analytics*. However, the account is still incomplete in several important aspects:

(A) We have not yet seen how the relevant *explananda* are to be determined. Are these fixed by resources independent of our practices of explanation? Does Aristotle use non-explanatory resources to fix (e.g.) the relevant primitive universals?

(B) We have not yet discovered the range of cases to which this model applies. How wide-ranging is the set of cases where the cause is separate from its effect? Can the model apply to kinds and substances?

Both (A) and (B) highlight areas where Aristotle's attempts to satisfy his own Definitional Constraint remain unclear. (A) is particularly pressing since (in addition to the reasons given above) it points to an incompleteness in his discussion of his central set of examples.

The Hierarchical Conception generates further questions:

(C) How are the species related to the higher-order kinds? Can one species fall within a variety of genera? If not, why not? Does

[363] For future discussion of B.2, $90^a$ 14ff., see Ch. 10 Sect. 1.

Aristotle require that the properties which species possess are unique to that species?

(D) More generally, how is Aristotle's method of definition by genus and *differentia* connected to definition by causal essence of the type discussed in this Chapter. Are these different types of definition? If so, how can the notion of definition be dependent (in the ways suggested) on the account of explanation?

I shall examine questions (A), (C), and (D) in the next Chapter in considering Aristotle's further views of definition in *Posterior Analytics* B.13–15. Question (B) will be discussed in Chapters 10 and 11.